



ADVANCED SOC ANALYST CERTIFICATION PROGRAM

Learn Security. Build Detection Skills. Defend Real-World Systems.

SKILL IT Education presents a structured, practical career pathway into Security Operations — taking learners step-by-step from IT and networking foundations to SIEM monitoring, incident response and real-time threat detection. Every concept is reinforced through hands-on labs, real-world tools and industry-aligned SOC practices.

PROGRAM HIGHLIGHTS

- ✓ 5 Core SOC Engineering Modules
- ✓ 5+ Hands-On Labs & Capstone Simulation
- ✓ SIEM, Incident Response & Threat Hunting
- ✓ Industry-Aligned SOC Practices

YOUR SOC ANALYST JOURNEY



WHY THIS MATTERS

Cyber threats now target every organization, and businesses across every industry depend on skilled SOC analysts to monitor, detect and respond before damage is done — making this one of the fastest-growing and most valuable career paths in technology today.

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Course Prepared by
IITian & Security Architect

Duration: 5 Months

3 Months Learning + 2 Months Internship

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01 Program Overview

Build Your SOC Analyst Career

What This Program Covers

This program takes you step-by-step from strong IT and networking foundations to real Security Operations Center work — monitoring networks, operating SIEM platforms, investigating incidents and hunting threats in live, hands-on environments.

Why This Matters

Cyber attacks now target every industry, and organizations urgently need engineers who can monitor, detect and respond — not just install security tools. SOC skills are among the highest in demand across the global technology job market.

COURSE DURATION

3 Months

PROJECT LEARNING

Integrated

INTERNSHIP

2 Months

PROGRAM ROADMAP

- 1 Foundations: IT, Networking & Operating Systems
- 2 The Cyber Threat Landscape
- 3 Inside a Security Operations Center
- 4 SIEM Platforms & Security Monitoring
- 5 Incident Response & Threat Hunting

*From Learning to Monitoring. From Alerts to Action.
From Skills to Career.*

02 Module 1 — Foundations: IT, Networking & OS

WHAT YOU WILL LEARN

- IT infrastructure & software fundamentals
- Networking essentials: LAN, MAN, WAN & Internet
- Network topologies, devices & the OSI/TCP-IP models
- IP addressing, classifications & subnetting
- Routers, switches, firewalls & IDS/IPS placement
- Windows & Linux installation and administration
- Linux terminal / CLI & command-line practice
- Reputation checks for suspicious IPs & URLs

TOOLS & TECHNOLOGIES

Windows Server

Linux / Ubuntu

CLI

Wireshark

Nmap

VirtualBox

NETWORKING STACK

IP

Subnet

Router

Switch

Firewall

IDS/IPS

✓ LEARNING OUTCOME

Students build a strong technical foundation for security work by configuring networks, working across Windows and Linux, and understanding how IT infrastructure actually operates.

03 Module 2 — The Cyber Threat Landscape

WHAT YOU WILL LEARN

- Threat, vulnerability & risk — how they connect
- CIA Triad and AAA security models
- Ethical hacking vs. hacking, and types of hackers
- The five phases of hacking: recon to clearing tracks
- Malware types: viruses, worms, trojans & ransomware
- Phishing, spear-phishing & business email compromise
- Password attacks, MITM, DoS & DDoS attacks
- Web application attacks: injection, XSS & more

TOOLS & TECHNOLOGIES



DEFENSIVE MEASURES



✓ LEARNING OUTCOME

Students learn to recognize, classify and understand the mechanics of common cyber attacks, building the attacker's-eye view every SOC analyst needs.

04 Module 3 — Inside a Security Operations Center

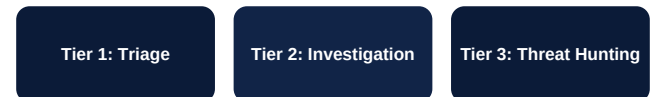
WHAT YOU WILL LEARN

- What a SOC does, and why it matters
- Key SOC functions: triage, investigation & hunting
- How a SOC is structured and staffed
- Key SOC roles and responsibilities
- Modern-day SOC: people, process & technology
- SOC as a Service (SOCaaS) and delivery models
- NOC vs. SOC — how the two teams differ
- Cyber security monitoring essentials

TOOLS & TECHNOLOGIES



SOC TIER STRUCTURE



✓ LEARNING OUTCOME

Students understand how a SOC is structured and staffed, and can explain the tools, roles and workflows that keep a security team running around the clock.

05 Module 4 — SIEM Platforms & Security Monitoring

WHAT YOU WILL LEARN

- SIEM architecture, event & flow collectors
- IBM QRadar: dashboards & offense analysis
- Malicious IP communication & phishing analysis
- Windows, Linux & security device log types
- Correlation rules, false positives & alert triaging
- Reference sets, use-case creation & rule tuning
- Log source integration across the enterprise
- Generating governance & security reports

TOOLS & TECHNOLOGIES

IBM QRadar

Splunk

Win Collect

Firewall Logs

IPS / WAF

Proxy Logs

SIEM WORKFLOW

Collect

Correlate

Detect

Investigate

Report

✓ LEARNING OUTCOME

Students learn to operate a SIEM platform end to end — building dashboards, investigating offenses, tuning rules and producing real security reports.

06 Module 5 — Incident Response & Threat Hunting

WHAT YOU WILL LEARN

- Incident handling process & response lifecycle
- Preparation, containment, eradication & recovery
- Security incident playbooks vs. runbooks
- The Cyber Kill Chain and MITRE ATT&CK framework
- Cyber Threat Intelligence (CTI) and its lifecycle
- Indicators of Compromise (IOC) vs. Indicators of Attack
- Threat hunting methods & investigation tools
- Email header analysis & malware analysis basics

TOOLS & TECHNOLOGIES

MITRE ATT&CK

Kill Chain

CyberChef

Sysinternals

Google Dorks

IOC / IOA

INCIDENT RESPONSE LIFECYCLE

Prep

Identify

Contain

Eradicate

Recover

Review

✓ LEARNING OUTCOME

Students learn to respond to real incidents using a structured lifecycle, and to hunt proactively using threat intelligence and industry-standard frameworks.

07 SOC Technology Stack

The complete toolset used across every module of the program

Networking

- Routers & Switches
- Firewalls
- IDS / IPS

Operating Systems

- Windows Server
- Linux / Ubuntu
- CLI Tools

SIEM Platforms

- IBM QRadar
- Splunk
- Win Collect Agent

Security Monitoring

- SOAR
- EDR / XDR
- DLP & IAM

Threat Intelligence

- MITRE ATT&CK
- Cyber Kill Chain
- IOC / IOA Feeds

Investigation Tools

- CyberChef
- Sysinternals
- Google Dorks

08 Hands-On Labs & Project-Based Learning

Build what you learn — every module is reinforced with hands-on work

01 Network & Lab Environment Setup

Configure IP addressing, subnetting and a virtual lab with Windows and Linux systems.

02 SIEM Monitoring Lab

Onboard log sources into IBM QRadar, create reference sets and build real-time dashboards.

03 Threat Detection Exercise

Analyze malware, phishing emails and malicious IP activity using live SIEM offense data.

04 Incident Response Simulation

Run through a full incident lifecycle — from identification to containment and recovery.

05 Threat Hunting Project

Use MITRE ATT&CK and threat intelligence feeds to proactively hunt for hidden threats.

FINAL CAPSTONE

End-to-End SOC Simulation

Monitor, investigate and respond to a simulated multi-stage attack from detection to resolution.

09 Learning Outcomes

What you will be able to do by the end of the program

- ✓ Configure and troubleshoot enterprise networks
- ✓ Operate SIEM platforms such as IBM QRadar & Splunk
- ✓ Recognize and classify common cyber attacks
- ✓ Monitor, triage and escalate security alerts
- ✓ Follow a structured incident response lifecycle
- ✓ Conduct proactive threat hunting
- ✓ Administer Windows Server and Linux environments
- ✓ Read and correlate security device logs
- ✓ Perform basic malware & email analysis
- ✓ Apply the Cyber Kill Chain and MITRE ATT&CK
- ✓ Use threat intelligence to strengthen detection
- ✓ Communicate findings through reports & playbooks

10 Career Opportunities

Build your career in Security Operations

SOC Operations

- SOC Analyst (L1)
- SOC Analyst (L2)
- Security Analyst

Incident & Threat Response

- Incident Response Analyst
- Threat Intelligence Analyst
- Junior Threat Hunter

Security Engineering

- SIEM / Security Engineer
- Vulnerability Analyst
- Network Security Engineer

Monitoring & Detection

- Security Monitoring Analyst
- Alert Triage Specialist
- Log Analysis Engineer

Compliance & Risk

- Cyber Risk Analyst
- IT Security Auditor
- Compliance Analyst

Advanced Career Paths

- SOC Team Lead
- Security Architect
- CISO (long-term path)

Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.



WHY CHOOSE SKILL IT EDUCATION

DON'T JUST LEARN SECURITY. LEARN TO DEFEND IT.

Learn. Monitor. Detect. Respond. Defend.

SKILL IT Education builds SOC analysts, not just certificate holders. Through an industry-driven curriculum, hands-on labs and real SIEM tools, every student graduates with practical monitoring skills, verified experience and the confidence to walk into a security operations role from day one.

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Let's build a safer future.

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